

US Patent & Trademark Office

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United States Patent Application Publication

20250276546

Kind Code

A1

Publication Date

September 04, 2025

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HYBRID CORD REINFORCEMENT FOR A TIRE

Abstract

The present invention is directed to a cord reinforcement for a tire comprising parallel cords wherein at least one of the cords is a hybrid cord comprising a flat monofilament and a multifilament yarn, wherein the flat monofilament comprises polyamide and the multifilament yarn is wrapped around the flat monofilament and comprises one or more of aramid, glass fiber material and carbon fiber material. The flat monofilament has a linear density within a range of 2200 dtex to 6000 dtex and the multifilament yarn has a linear density within a range of 1800 dtex to 4400 dtex. Furthermore, the invention is directed to a tire comprising one or more of such cords.

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Family ID: 94768631

Appl. No.: 18/592979

Filed: March 01, 2024

Publication Classification

Int. Cl.: B60C9/00 (20060101); D02G3/04 (20060101); D02G3/38 (20060101); D02G3/48 (20060101)

U.S. Cl.:

CPC B60C9/005 (20130101); B60C9/0042 (20130101); B60C9/0064 (20130101); D02G3/045 (20130101); D02G3/047 (20130101); D02G3/38 (20130101); D02G3/48 (20130101); B60C2009/0071 (20130101)

Background/Summary

FIELD OF THE INVENTION

[0001] The present invention is directed to a cord reinforcement for a tire, wherein the cord reinforcement comprises hybrid cords. Furthermore, the invention is directed to a tire comprising one or more of such hybrid cords.

BACKGROUND OF THE INVENTION

[0002] Multiple rubber components of tires comprise cord reinforcements. Such cord reinforced rubber components help to provide robust and comfortable tires. Furthermore, in some applications, such as in truck and off-the-road tire applications, there is a specific demand for durable and/or cut resistant tire constructions. While developments have been made in this field over the past years, significant room for improvement remains.

SUMMARY OF THE INVENTION

[0003] In a first aspect, the present invention is directed to a cord reinforcement for a tire, comprising parallel cords wherein at least one of the cords is a hybrid cord comprising a flat monofilament and a multifilament yarn, wherein the flat monofilament comprises polyamide and the multifilament yarn is i) wrapped around the flat monofilament and ii) comprises one or more of aramid, glass fiber material and carbon fiber material. The flat monofilament has a linear density within a range of 2200 dtex to 6000 dtex and the multifilament yarn has a linear density within a range of 1800 dtex to 4400 dtex.

[0004] In a second aspect, the present invention is directed to a tire comprising at least one rubber component reinforced by at least one hybrid cord comprising a flat monofilament and a multifilament yarn, wherein the flat monofilament comprises polyamide and the multifilament yarn is wrapped around the flat monofilament and comprises one or more of aramid, glass fiber material and carbon fiber material. The flat monofilament has a linear density within a range of 2200 dtex to 6000 dtex and the multifilament yarn has a linear density within a range of 1800 dtex to 4400 dtex.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The invention will be described by way of example and with reference to the accompanying drawings in which:

[0006] FIG. 1 is a schematic cross-section of a pneumatic tire comprising a carcass ply reinforced by hybrid cords in accordance with an embodiment of the present invention;

[0007] FIG. 2 is a schematic partial cross section of a carcass ply reinforced by parallel hybrid cords in accordance with an embodiment of the present invention;

[0008] FIG. 3 is a schematic cross section of a hybrid cord, such as already shown in FIG. 2; and

[0009] FIG. 4 is a schematic partial perspective view of a fabric comprising multiple hybrid cords, such as those shown in FIGS. 2 and 3, also in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0010] According to the first aspect, the cord reinforcement for a tire comprises parallel cords wherein at least one of the cords is a hybrid cord comprising a flat monofilament and a multifilament yarn, wherein the flat monofilament comprises polyamide and the multifilament yarn is i) wrapped around the flat monofilament and ii) comprises one or more of aramid, glass fiber material and carbon fiber material. The flat monofilament has a linear density within a range of 2200 dtex to 6000 dtex and the multifilament yarn has a linear density within a range of 1800 dtex to 4400 dtex.

[0011] In one embodiment, the multifilament yarn is helically wrapped around the flat monofilament. In particular, the multifilament yarn is not twisted together with the flat

monofilament. In other words, the monofilament can optionally be described as a core filament, or core monofilament, of the cord, which is preferably untwisted.

[0012] In another embodiment, most of or all of the parallel cords are said hybrid cords comprising the flat monofilament and the multifilament yarn.

[0013] In another embodiment, the flat monofilament has a linear density within a range of 4000 dtex to 5000 dtex, and/or the multifilament yarn has a linear density within a range of 1800 dtex to 2600 dtex.

[0014] In still another embodiment, the linear density of the flat monofilament is at least 10%, preferably at least 20%, or even more preferably at least 50% higher than the linear density of the multifilament yarn. In addition, or alternatively, the linear density of the flat monofilament is at most 300%, or preferably at most 250%, or even more preferably at most 150% higher than the linear density of the multifilament yarn.

[0015] In still another embodiment, the flat monofilament has one or more of i) a width within a range of 1.0 mm to 1.3 mm; and ii) a thickness within a range of 0.3 mm to 0.6 mm.

[0016] The term flat describes herein the form of the monofilament in a cross-section perpendicular to an elongated extension of the monofilament. The thickness and the width are also measured herein in such a cross-section. Optionally, the flat monofilament is a monofilament having an aspect ratio of width over thickness (in other words, width divided by thickness) within a range of 1.5 to 4.5, preferably within a range of 2 to 4. In case the thickness and/or width is not constant, e.g., along the elongated extension of the monofilament, the largest thickness and/or width is considered herein. The width is measured perpendicular to the thickness and the width is chosen as being larger than the thickness.

[0017] In still another embodiment, the flat monofilament comprises polyamide 6.6 (PA 6.6), such as Nylon™ 6.6.

[0018] In still another embodiment, the flat monofilament has an essentially elliptical cross section, or in other words cross-sectional form. Elliptical cross sections have turned out to allow a better winding of the multifilament yarn around the flat monofilament and/or avoid damages to the multifilament yarn upon winding. At the same time, a relatively low thickness is provided which allows for use of less rubber composition material in tire plies, particularly in tire carcass plies.

[0019] In still another embodiment, the multifilament yarn has one or more of a linear density within a range of 1800 dtex to 2600 dtex; and a diameter within a range of 0.2 mm to 0.6 mm, preferably of 0.2 mm to 0.5 mm.

[0020] In still another embodiment, the multifilament yarn comprises from 2 to 3 filaments, preferably 2 filaments, which are twisted together to form the multifilament yarn.

[0021] In still another embodiment, the multifilament yarn has a lay length in the cord, which is within a range of 1 mm to 30 mm, preferably 2 mm to 20 mm, or even more preferably 3 mm to 10 mm. Lay length means herein a length, or in other words a distance, of one (e.g., helical) revolution of the multifilament yarn along the monofilament, or equivalent. The multifilament yarn may be described as sheath (multifilament) yarn of the cord and the monofilament may be described as core monofilament of the cord so that the core monofilament and the sheath yarn form the cord.

[0022] In still another embodiment, the multifilament yarn comprises 2 aramid filaments twisted together to form the multifilament yarn.

[0023] In still another embodiment, each of the aramid filaments is (individually) twisted with 50 turns per meter (TPM) to 350 TPM. In other words, each of the aramid filaments as such is twisted within such a range. In addition, or alternatively, one of the aramid filaments is (individually) twisted in S-direction and another one of the aramid filaments is (individually) twisted in Z-direction, such as within the TPM range mentioned above.

[0024] In still another embodiment, both aramid filaments are twisted together within a range of 100 TPM and 400 TPM to form the multifilament yarn. Thus, optionally, each of the aramid filaments may be individually twisted as mentioned in the preceding paragraph, and then again

twisted within the range of 100 TPM and 400 TPM to form the multifilament yarn.

[0025] In still another embodiment, the hybrid cord has a breaking strength within a range of 350 N to 700 N, preferably 400 N to 600 N, determined according to ASTM D885, or equivalent.

[0026] In still another embodiment, the hybrid cord has a maximum thickness within a range of 0.6 mm to 0.9 mm. The thickness of the hybrid cord (including the flat monofilament and the multifilament yarn) is determined perpendicular to the width (mentioned in some embodiments as w) of the flat monofilament.

[0027] In still another embodiment, the hybrid cord is a textile hybrid cord.

[0028] In still another embodiment, the cord reinforcement is a fabric, preferably a textile fabric.

[0029] In still another embodiment, said fabric is a woven fabric. Optionally, the fabric comprises warp cords and weft cords. Preferably, said hybrid cords are the warp cords. Preferably, the weft cords have a linear density which is at least five times lower than the linear density of the hybrid cords, and/or the warp cords, respectively. As typical in the art, linear density means herein a weight per length, such as of a filament, yarn, or cord. For instance, it can be provided in the unit dtex as typical in the art.

[0030] According to the second aspect, a tire comprises at least one rubber component reinforced by one or more hybrid cords wherein each of the hybrid cords comprises a flat monofilament and a multifilament yarn, wherein the flat monofilament comprises (or consists of) polyamide and the multifilament yarn is wrapped around the flat monofilament and comprises (or consists of) one or more of aramid, glass fiber material and carbon fiber material. The flat monofilament has a linear density within a range of 2200 dtex to 6000 dtex and the multifilament yarn has a linear density within a range of 1800 dtex to 4400 dtex.

[0031] In one embodiment, said rubber component is one of a belt ply, a carcass ply, an overlay ply, and a ply strip. A ply is understood herein as a layer comprising a rubber composition reinforced by cords and/or a fabric. Rubber compositions of such rubber components or plies are known in the art as such.

[0032] In another embodiment, the rubber component is a carcass ply, preferably an outermost carcass ply. For instance, if the tire has multiple carcass plies, such as two or three carcass plies, e.g. extending along one another in a sidewall region of the tire, the outermost carcass ply is the carcass ply which is outermost with respect to an equatorial plane of the tire in said sidewall region of the tire.

[0033] In still another embodiment, the tire is a pneumatic tire comprising a tread portion, a pair of axially spaced sidewalls, a pair of axially spaced bead portions, one or more, preferably, one or two, (cord reinforced) carcass plies connecting both bead portions and extending along (and e.g., optionally through) each sidewall of the pair of axially spaced sidewalls, and at least one belt arranged radially between the tread portion and the one or more cord reinforced carcass plies in a crown region of the tire, wherein said rubber component is, preferably, a carcass ply of the one or more cord reinforced carcass plies.

[0034] In still another embodiment, said carcass ply comprises multiple of the hybrid cords comprising the flat monofilament and the multifilament yarn, and/or the hybrid cords are arranged in parallel to one another.

[0035] In still another embodiment, the carcass ply has one or more of a thickness, measured in an equatorial plane of the tire, within a range of 1.4 mm to 1.9 mm, preferably of 1.5 mm to 1.8 mm, and/or has an ends per inch (EPI) range from 10 EPI to 14 EPI. In other words, said carcass ply comprises from 10 cords per inch to 14 cords per inch, determined along a direction perpendicular to the parallel extension of the cords.

[0036] In still another embodiment, the tire is one of a radial pneumatic truck tire, and a radial pneumatic passenger car tire. While these are preferred embodiments, other tire applications are possible. For instance, the tire may also be a race tire or an aircraft tire.

[0037] In another aspect of the present invention, a tire comprises the cord reinforcement according

to the first aspect of the invention or one of its embodiments.

[0038] FIG. 1 shows a pneumatic radial tire **1** in accordance with an embodiment of the present invention. The tire **1** comprises a pair of axially spaced bead portions **2**, a tread portion **10**, a carcass ply **3**, which is folded around each bead **4** of the respective bead portion **2** and connects both bead portions **2** along a pair of sidewalls **6** of the tire **1**. Two belts **8, 9** or in other words, two belt plies, are arranged radially between the carcass ply **3** and the tread portion **10** in a crown region of the tire **1**. The carcass ply **3** comprises a cord reinforcement in accordance with an embodiment of the present invention, such as indicated in FIG. 2.

[0039] For the sake of better understanding the radial direction r , the axial direction a , and the circumferential direction c are indicated in FIG. 1. The axial direction a is parallel to the axis of rotation of the tire **1**. The radial direction r is perpendicular to the axial direction a . The circumferential direction c is perpendicular to the to the axial direction and the radial direction, and, e.g., parallel to a circumferential centerline of the tire **1** or its equatorial plane EP. References to one of these directions are not necessarily limited to a specific orientation, unless indicated otherwise herein.

[0040] While FIG. 1 shows only two belts **8, 9** and one carcass ply **3**, other embodiments include, e.g., three or four belts, and/or two carcass plies, such as in some light truck, off-the-road, and truck tires.

[0041] FIG. 2 depicts a schematic partial cross-section of the carcass ply **3** of FIG. 1, which comprises a rubber composition **34** (preferably a sulfur-cured rubber composition) reinforced by a plurality of parallel hybrid cords **30**. The gauge g , or in other words thickness, of the carcass ply **3**, is measured perpendicular to the carcass ply **3**, such as in the equatorial plane EP of the above-mentioned tire **1**. Preferably, such a gauge g is within a range of 1.4 mm to 1.9 mm for the carcass ply **3** in accordance with the present embodiment of the invention.

[0042] FIG. 3 schematically shows a hybrid cord **30** in further details. The hybrid cord **30** comprises a flat monofilament **31**, here with a preferred elliptical cross-section having the width w and the thickness t , wherein the flat monofilament **31** of the hybrid cord **30** is wrapped with a sheath multifilament yarn **32**. In the preferred but non-limiting embodiment, this multifilament yarn **32** comprises two single (mono)filaments **23** which are twisted with one another (or together) to form the multifilament yarn **32**. Moreover, each filament **23** is twisted as such. In particular, a first filament **23** of the multifilament yarn **32** is twisted in S-direction and a second filament **23** of the multifilament yarn **32** is twisted in Z-direction, and both filaments **23** of the multifilament yarn **32** are twisted together to form the multifilament yarn **32**. Finally, the multifilament yarn **32** is helically twisted or wrapped around the flat (core) monofilament. Thus, in the present embodiment a textile hybrid cord **30** is formed. Optionally, one or more filaments, the yarn and/or the cord are coated and/or dipped, such as with one or more coatings and/or one or more dips, e.g., for improved adhesion to rubber composition material. Such coatings and/or dips are known in the art, such as resorcinol formaldehyde latex (RFL) dips. The same applies to rubber compositions, such as those used in the carcass ply, and/or other rubber components of the tire.

[0043] One preferred construction of the multifilament yarn is 1100/2 aramid 300S/300Z/350 TPM. The final twist of the two aramid filaments to form the yarn may either be in S or Z direction. Another preferred construction is 1670/2 dtex aramid 100S/100Z/200 TPM. Again, the final twist may be in S or Z direction. Yet another preferred construction is 1100/2 aramid 100S/100Z/200 TPM, with the final twist in S or Z direction. The monofilament yarn is in one non-limiting embodiment a 4400/1 dtex Nylon 6.6 flat monofilament. It does not have a twist. The multifilament yarn preferably has a lay length of about 3 mm to 10 mm, and/or is wound around the flat monofilament with 100 to 300 TPM. Preferred diameter, thickness, and/or width ranges have already been provided herein above.

[0044] FIG. 4 shows a schematic partial perspective view of a cord reinforcement for a tire, which is a textile fabric **300** comprising a plurality of parallel hybrid cords **30**, wherein each of the cords

30 is formed by a flat monofilament **31** and a multifilament yarn **32** wrapped around the monofilament **31** (as already shown in FIG. 3). In the present embodiment, the parallel hybrid cords **30** are warp cords connected by weft cords **35**, preferably woven with the hybrid cords **30**, or in other words the warp cords. The weft cords may be mono filament cords or multifilament cords. Their linear density is typically lower than the linear density of the warp cords. In a carcass ply of a tire, a hybrid and/or warp cord of such a textile fabric preferably extends in a plane perpendicular to the equatorial plane of the tire.

[0045] In particular, cord reinforcements including hybrid cords with said flat monofilament and wrapped by said multifilament yarn have turned out to provide an advanced cut resistance and/or durability, particularly when used as cord reinforcement in at least one carcass ply of the tire. The use of polyamide, such as PA 6.6, as flat core monofilament wrapped by the multifilament yarn within the mentioned linear density ranges further provides a favorable balance between elongation and breaking strength of the cord, e.g., particularly an elongation provided by the polyamide filament and breaking strength provided by the sheath yarn, e.g., made of aramid.

[0046] All aspects, embodiments, and features, which are mentioned herein, can be combined with one another.

[0047] Variations in the present invention are possible in light of the description of it provided herein. While certain representative embodiments and details have been shown for the purpose of illustrating the subject invention, it will be apparent to those skilled in this art that various changes and modifications can be made therein without departing from the scope of the subject invention. It is, therefore, to be understood that changes can be made in the particular embodiments described which will be within the full intended scope of the invention as defined by the following appended claims.

Claims

1. A cord reinforcement for a tire comprising parallel cords wherein at least one of the cords is a hybrid cord comprising a flat monofilament and a multifilament yarn, wherein the flat monofilament comprises polyamide and the multifilament yarn is i) wrapped around the flat monofilament and ii) comprises one or more of aramid, glass fiber material and carbon fiber material, and wherein the flat monofilament has a linear density within a range of 2200 dtex to 6000 dtex and the multifilament yarn has a linear density within a range of 1800 dtex to 4400 dtex.
2. The cord reinforcement according to claim 1, wherein the multifilament yarn is helically wrapped around the flat monofilament.
3. The cord reinforcement according to claim 1, wherein the flat monofilament has a linear density within a range of 4000 dtex to 5000 dtex, and the multifilament yarn has a linear density within a range of 1800 dtex to 2600 dtex.
4. The cord reinforcement according to claim 1, wherein the flat monofilament has one or more of: a linear density which is at least 10% higher than a linear density of the multifilament yarn; a maximum width of 1.0 mm to 1.3 mm; and a maximum thickness of 0.3 mm to 0.6 mm.
5. The cord reinforcement according to claim 1, wherein the flat monofilament comprises polyamide 6.6.
6. The cord reinforcement according to claim 1, wherein the flat monofilament has an essentially elliptical cross section.
7. The cord reinforcement according to claim 1, wherein the multifilament yarn has one or more of: a linear density within a range of 1800 dtex to 2600 dtex; and a diameter within a range of 0.2 mm to 0.6 mm.
8. The cord reinforcement according to claim 1, wherein the multifilament yarn comprises from 2 to 3 filaments twisted together to form the multifilament yarn.
9. The cord reinforcement according to claim 1, wherein the multifilament yarn has a lay length in

the cord, which is within a range of 3 mm to 10 mm.

10. The cord reinforcement according to claim 1, wherein the multifilament yarn has 2 aramid filaments twisted together to form the multifilament yarn.

11. The cord reinforcement according to claim 10, further with one or more of: each of the aramid filaments being individually twisted with 50 TPM to 350 TPM; one of the aramid filaments being individually twisted in S-direction and another one of the aramid filaments being individually twisted in Z-direction; and both aramid filaments being twisted together within a range of 100 TPM and 400 TPM to form the multifilament yarn.

12. The cord reinforcement according to claim 1, wherein the hybrid cord has a breaking strength within a range of 350 N to 700 N.

13. The cord reinforcement of claim 1, wherein the hybrid cord has a maximum thickness within a range of 0.6 mm to 0.9 mm.

14. A tire comprising at least one rubber component reinforced by one or more hybrid cords, each of the hybrid cords comprising a flat monofilament and a multifilament yarn, wherein the flat monofilament comprises polyamide and the multifilament yarn is wrapped around the flat monofilament and comprises one or more of aramid, glass fiber material and carbon fiber material, and wherein the flat monofilament has a linear density within a range of 2200 dtex to 6000 dtex and the multifilament yarn has a linear density within a range of 1800 dtex to 4400 dtex.

15. The tire according to claim 14, wherein said rubber component is one of a belt ply, a carcass ply, an overlay ply, and a ply strip.

16. The tire according to claim 14, wherein the rubber component is a carcass ply.

17. The tire according to claim 14, wherein the tire is a pneumatic tire comprising a tread portion, a pair of axially spaced sidewalls, a pair of axially spaced bead portions, one or more cord reinforced carcass plies connecting both bead portions and extending along each sidewall of the pair of axially spaced sidewalls, and at least one belt arranged radially between the tread portion and the one or more cord reinforced carcass plies in a crown region of the tire, and wherein the rubber component is a carcass ply of the one or more cord reinforced carcass plies.

18. The tire according to claim 17, wherein the carcass ply comprises multiple of the hybrid cords comprising the flat monofilament and the multifilament yarn, and wherein the hybrid cords are arranged in parallel to one another.

19. The tire according to claim 18, wherein the carcass ply has one or more of: a thickness, measured in an equatorial plane of the tire, within a range of 1.4 mm to 1.9 mm; an EPI range from 10 EPI to 14 EPI.

20. The tire according to claim 14, wherein the tire is a one of a radial pneumatic truck tire, and a radial pneumatic passenger car tire.
